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Silicon wafer defect pattern detection using machine learning

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ABSTRACT

A silicon integrated circuit has been created on silicon wafers by diffusing doping materials on the surface of the same. The silicon wafer is a thin disk made from semiconductor material (mostly ultrapure silicon). Silicon chip fabrication is an industrial process; hence defects occur on silicon wafers caused by unclean factory environment, production machine errors and fabrication process errors. The accurate classification of wafer defects provides information for engineers through regular patterns in the processes. Manual inspection of every chip is not economically feasible. In this study, WM-811k is used as the dataset. This is one of the most extensive wafer bin map dataset available open source on kaggle. It was created by MIRLABS. The dataset was compiled by taking the test data on 811,457 silicon wafer, in which each wafer map image was collected from real-world fabrication. A CNN and DNN model is used to classify the defect patterns.

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1. Introduction

The process of creation of modern integrated circuits (ICs) is highly automated and uses properly controlled equipment and environment. As nothing is perfect, machine malfunction and process drift still occur. The process of dicing the individual ICs for packaging also creates imperfections in them. This is unavoidable due to the complexity of hundreds of steps in operation. Some of the large number of steps includes diffusion, oxidation, etching, and deposition. Small errors in a single step can cause a large error in the doping accuracy of each IC on the silicon wafer and can cause considerable IC failures. Semi-conductor fabrication industry aims to improve the productivity to max by taking measures against factors that limit the yield. Controlling the ratio of the working ICs to the defective ones represent the amount of control a FAB has over the process [1]. Defects induced by first few steps in the fabrication of the IC reflect the flaws in the manufacturing equipment and the process. These are represented as random defects and systematic ones based on the original causes [2].

Random errors caused by the failure of manufacturing equipment and environment are not easy to classify. Elimination of these requires very expensive maintenance of the equipment and even

more expensive environment control measures. Even in an almost sterile environment, it is not possible to guaranty the absence of dust particles. These occur in small number, still reducing the numbers does improve the yield of the foundry [3]. Systematic defects are regular and predictable in nature; their occurrence may be related to some specific manufacturing process. These defects can be reduced by modifying the part of fabrication process related to them. These occur in regular intervals forming a shape or a locus of failures related to the original causes. The analysis of these defects helps in better engineering of manufacturing processes; this improves the die quality, reduces failure rate, and increases process yield [4]. Defect recognition targets systematic ones only. Defects that are random in nature are considered as noise as they cause errors in the process of detection of the systematic defects.

All the ICs have to be tested before they are packaged to reduce waste (and loss). Trained engineers may test a few, but in the volumes of current production, that is not practical. Hence, most of the testing in the industry is done through automation. Defects are studied by the visualization of wafer maps. The patterns that are found in the wafer map visualization is important for the classification. The patterns that are detected by an AI may also help in linking the failures to the intricacies of the fabrication process. By analyzing the cause of a particular problem on the production line related to a defect type, solutions to reduce such failure may be

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developed. Since the process is automatic, it is very economical to use in the industry.

Accuracy cost and time effective silicon wafer tools are high in demand in the industry. Deep Neural Networks are more accurate due to the multilayer perception. The accuracy of a neural network can be increased by increasing the number of layers. The same can also be applied to convolutional Neural Networks. This research uses multiple convolution layers followed by a block of deep neural network. By using CNN + DNN, the achieved accuracy of the model is 99.45%. The following is the arrangement of this paper:

Section 2 discusses the recent progress in this field. Section 3 annotates the concepts used in the research. Section 4 explores the dataset and the model proposed. Section 5 gives the detailed experiment and its results. Section 6 describes conclusion and references.

2. Related work

There have been significant developments in the Machine Learning (ML) based solutions for wafer map defect analysis. The Convolutional Neural Networks (CNN) have been used, comprised of a number of convolution and layers of pooling, succeeded by a densely connected layer; Finally there is the output (soft-max) layer that generates a probability of the classification [5]. The Systematic Literature Review (SLR) is published to discuss how the silicon IC manufacturing enterprise is benefitted by deep learning research advancements for wafer map defect recognition and analysis [6]. CNN and image retrieval has been used to classify wafer defect patterns [7].

The double feature extraction method for CNN has been also used. First the radon transform (feature extraction) was used, that was then fed to a CNN (2nd feature extraction) [8]. New models with MobileNetV1 and V2 for two different datasets have been created to better classify the defects of the silicon wafer map. A simpler deep convolution architecture helps reduce the number of variables and calculation required to implement the model, which is more economic for the testing with large production volumes. After the data augmentation to reduce the imbalance, classification accuracy is improved to 97.01% and 95.09%, respectively [9]. The classification accuracy and the cost of creating a classifier have been improved. There are many bottlenecks in conventional technology, a deep CNN has been proposed based on the “VGG16 architecture”, and perform data processing on the image dataset. The proposed model was efficiently trained on a small amount of data (images) and achieved high accuracy of detecting defects [10]. The problem of data imbalance had been solved after applying image processing technique on the data that was used to train the model [11]. Datasets of wafer maps are few and far between, hence the difficulty of over-fitting has been solved by detecting wafer map defects using a model based on “Mobile-Net V2” [12]. The process of defect classification using Defect Database Management System (DDMS), developed in-house at SCL(India), has also been explored [13]. Self-supervised learning for the artificial intelligence (AI) models has been used to increase the efficiency of current pro-

cesses [14]. The use of GAN has improved the accuracy in detection of the wafer failure patterns. Generator of the GAN generates the wafer maps to reduce the imbalance the dataset [15]. ResNet101 when paired with multi-granularity GAN (MGGAN) achieved an accuracy of 93.43% [16]. Using a “convolutional variation auto encoder” (CVAE), a recall of 96.9% has been achieved [17].

3. Concepts

3.1. Wafer bin map

An integrated circuit (IC) is fabricated on disks of base semiconductor material called a silicon wafer. The architecture is the design of the IC. This is formed on the wafer by diffusing the doping elements on the wafer surface. Then formed ICs are tested each by the tester at the foundry. The Wafer Bin Map (WBM) is the representation of the working and the failed ICs on the particular wafer as shown in Fig. 1.

3.2. Machine learning

ML is a framework that emulates the intelligence found in self-aware beings. A model has weights similar to neurons that form a network similar to the human brain (many orders of magnitude simpler). The model starts without knowing the correct answer (prediction). During the process of training or fitting of a model over the dataset, the optimizing algorithm changes the model weights to better the accuracy of the model predictions.

3.3. Neural networks

Also called artificial neural networks (ANNs); these form the basic part of the artificial intelligence (AI) created, trained and tested. Artificial neural networks are composed of nodes that are interconnected in a network as shown in Fig. 2. The exact architecture of the artificial neural networks are based on the arrangement of the neurons in the brains of intelligent species. There are multiple layers of such neurons – Input-layer, hidden-layers and the output-layer.

3.4. Deep neural networks

An ANN that is comprised of multiple hidden densely connected layers is called a deep neural network. DNNs are also well suited to model complex non-linear relationships [13]. Deep ANNs use the hidden layers to store and evaluate the significance of one of the inputs is to the output. The hidden layer stores information on significance of the inputs and it also makes connections between the significance of combinations of inputs. This is called multi-layer perception. Deep neural nets then utilize the multi-layer perception to increase the classification accuracy. The increase in accuracy increases with the number of hidden layers. A typical DNN example is Fig. 3.

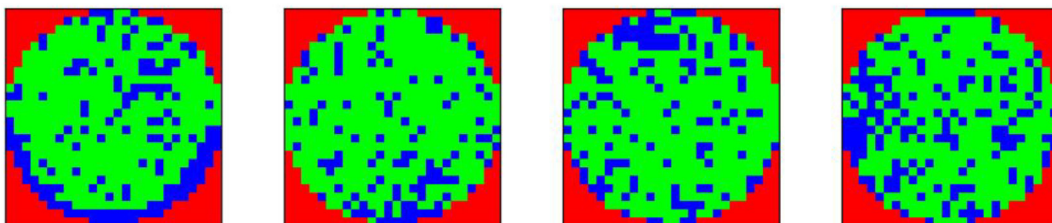


Fig. 1. Wafer bin map examples.

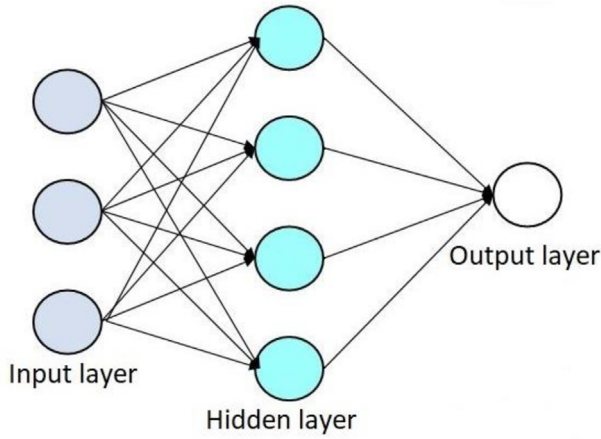


Fig. 2. Typical artificial neural network.

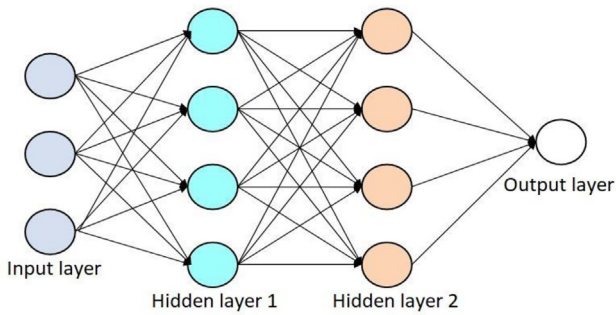


Fig. 3. Typical deep neural network.

3.5. Convolution layer

The building block of the CNN is the convolution layer. It can be seen as a collection of small square templates (convolutional kernels) that go over the image and look for the specific pattern (a trained CNN will). When the specific pattern is detected, the kernel

will return a large number. It will return a zero or small number for non-matching parts of the image. In the two-dimensional convolution (conv2d) kernel moves in both x, y dimensions.

3.6. Dense layer

The dense layer in a neural network is the layer that is deeply connected to the layers before it. It means that each neuron of the layer is getting its input from every neuron of the preceding layer. This is the most commonly used layer in the ANN. The neurons of the dense layer represent a matrix vector multiplication with the input data. The output of the same will be a high number if the trained pattern of data is detected. If not, the result will be a small number. In ANN, multiple layers of this type are used to get more accurate predictions.

4. Dataset and model

The dataset used is the WM-811k [14], also used in previous work done in this field. It comprises of 811,457 wafer maps (images), taken from 46,393 semiconductor wafer lots. This was created by MIR LABS and is available open source on kaggle. These are the types of defect patterns – “Center”, “Donut”, “Edge-Locus”, “Edge-Ring”, “Locus”, “Random”, “Scratch”, “Near full” and “None”. These are all seen in Fig. 4.

Other than the wafer maps, the dataset also has information that is not required. Hence it is removed. Fig. 5 shows the information.

This data set has many non-labeled images and many do not have a defect at all. Only 3.1% of the wafer maps are having both a label and pattern. Hence in this case only the images with a label can be used to develop the AI model. Fig. 6 shows the distribution of wafers in the WM-811k dataset.

4.1. CNN and DNN model architecture

The deep neural network achieves its better accuracy by the multilayer perception. When there are layers between the input and the output of a neural network that is deep, such that the abstraction causes their presence to not be observed by either input or output- such a network is called a deep neural network. These hidden layers store in their weights the data that represents

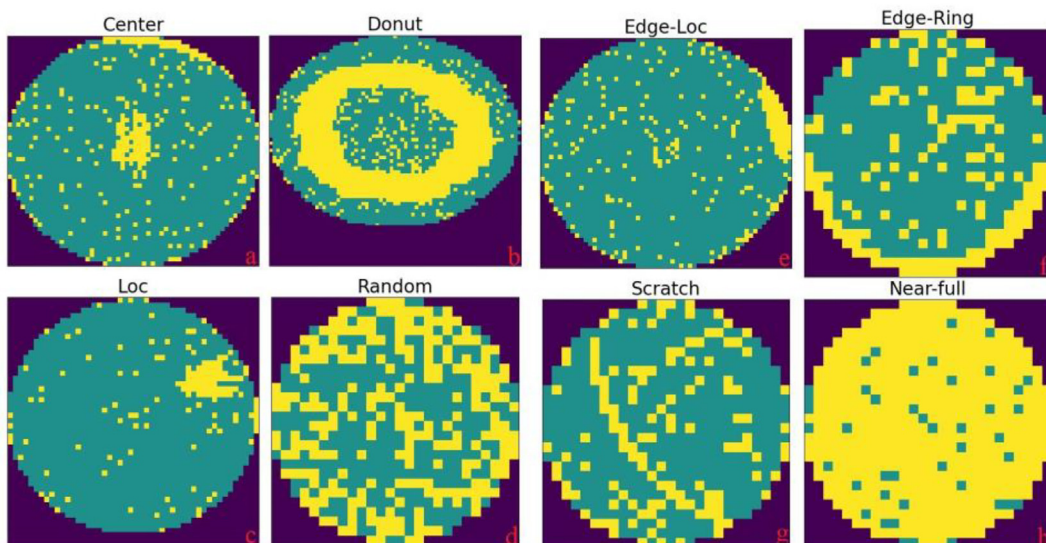


Fig. 4. Types of failures in WM-811k. (a) Center, (b) Donut, (c) Loc, (d) Random, (e) Edge Loc, (f) Edge Ring, (g) Scratch, (h) Near full.

	waferMap	dieSize	lotName	waferIndex	trianTestLabel	failureType
2280	[[0, 2, 2, 1, 1, 1,...	533.0	lot92	14.0	[[Training]]	[[none]]
2281	[[0, 1, 2, 1, 2, 1,...	533.0	lot92	15.0	[[Training]]	[[none]]
2282	[[0, 1, 1, 1, 1, 1,...	533.0	lot92	16.0	[[Training]]	[[none]]
2283	[[0, 2, 1, 1, 1, 1,...	533.0	lot92	17.0	[[Training]]	[[none]]
2284	[[0, 2, 2, 1, 1, 1,...	533.0	lot92	18.0	[[Training]]	[[none]]

Fig. 5. Some entries in the WM-811k.

the significance of the inputs to the outputs. They also store the associations between the combinations of the inputs.

The proposed methodology here is to use the multilayer perceptron in the convolutional neural networks. To further increase the accuracy, we add another block of deep neural network as shown in Fig. 7.

4.2. Proposed model

The model proposed here has a block of 5 layers deep of 1, 2-dimension convolution. This is followed by a flatten layer since convolution is a two-dimensional operation and DNN needs a linear form of data. This is followed by a 5-layer deep neural network of dense layers. The model summary is shown in Table 1.

5. Experiment and results

5.1. Training and validation

The data splits into parts for training, validation and testing. The training of the model was done on the training part of the dataset for a number of times (epochs). The model developed here was saved using the save functionality of the Tensor flow ("Keras") platform. Since the model can be saved and reloaded, the entire training process can be repeated multiple times on the same model. This also increases the accuracy of the model better than previous.

5.2. Testing

All machine learning, artificial intelligence models have to be properly tested for accuracy. The conventional manner for testing the accuracy of a model is to check its predictions on data that is different from the training data, which is called real world testing, and it represents the actual performance of an AI model. In semiconductor space, the amount of secrecy and trade secrets involved means that- WM-811k is the only dataset for such research, i.e. this is the only dataset available open source for this research. For the

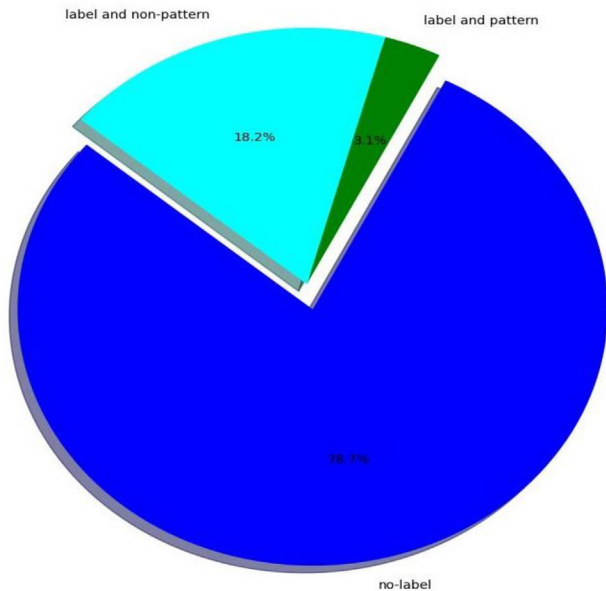


Fig. 6. Composition of the WBM in the WM-811k.

Table 1

Model Summary.

Layer (type)	Output Shape	Parameter #
input_1 (Input Layer)	[(None, 26, 26, 3)]	0
conv2d (Conv2D)	(None, 26, 26, 32)	896
conv2d_1 (Conv2D)	(None, 26, 26, 32)	9248
conv2d_2 (Conv2D)	(None, 26, 26, 32)	9248
conv1d (Conv1D)	(None, 26, 26, 32)	3104
conv2d_transpose (Conv2DTranspose)	(None, 28, 28, 24)	6936
flatten (Flatten)	(None, 18816)	0
dense (Dense)	(None, 32)	602,144
dense_1 (Dense)	(None, 32)	1056
dense_2 (Dense)	(None, 32)	1056
dense_3 (Dense)	(None, 32)	1056
dense_4 (Dense)	(None, 9)	297

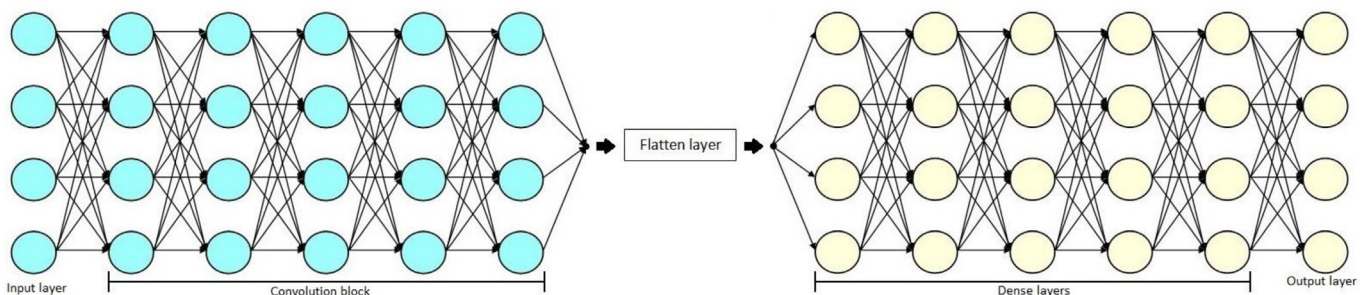


Fig. 7. Deep CNN and DNN Architecture.

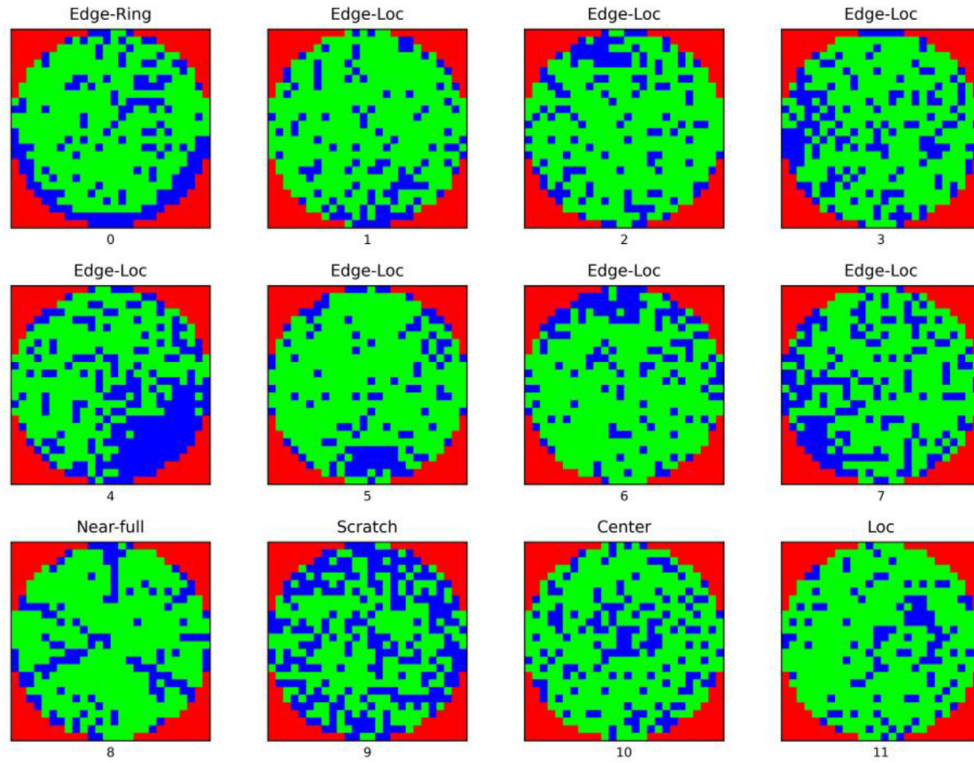


Fig. 8. Wafer Maps with predicted labels.

Table 2

Experimental result.

Train accuracy	Validate accuracy	Test accuracy
99.87%	99.55%	99.45%

Table 3

Result Comparisons.

Nakazawa et al. [7]	Shawon et al. [11]	This research Deep-CNN + DNN
98.20%	99.29%	99.45%

purpose of testing the accuracy of the model, 10% of the dataset had to be set aside for the testing of this model.

5.3. Results

The data was divided into three parts, which means there will be three different accuracy values in our result. This result had achieved after the model was trained, validated and tested for 35 epochs (times) as shown in Table 2.

Fig. 8 shows different wafer maps along with their prediction labels. Then we compare the results that have been published in the past with that of this research. By using the DNN along with

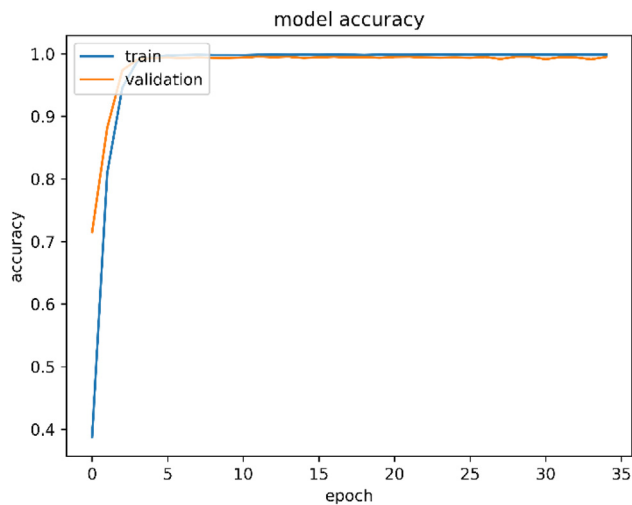


Fig. 9. Accuracy and the validation accuracy plot.

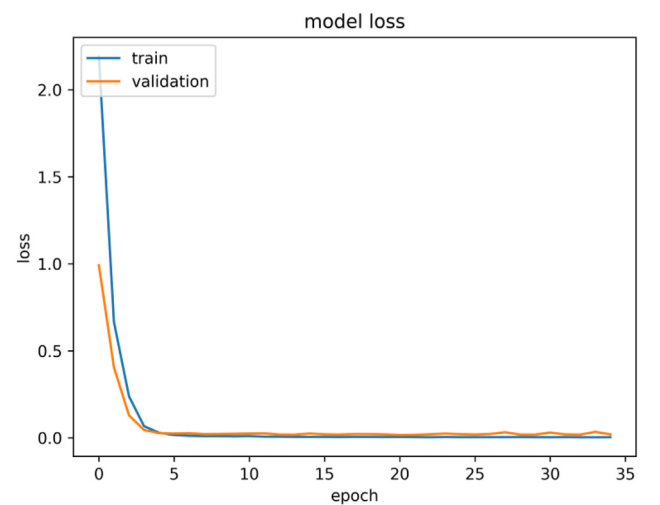


Fig. 10. Model loss plot.

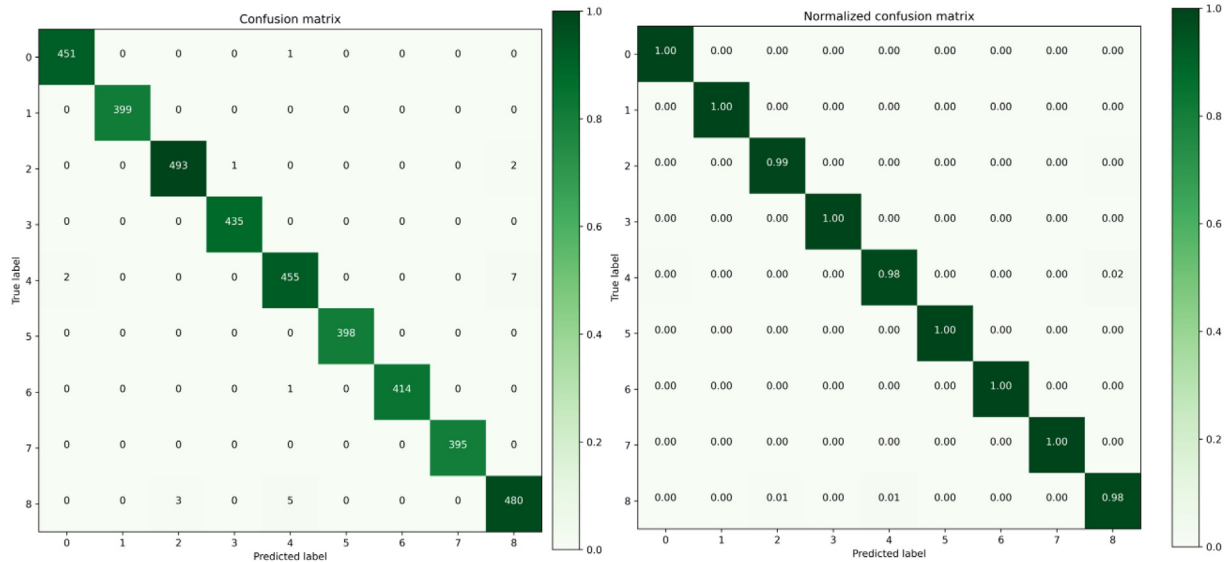


Fig. 11. Confusion Matrix.

a deep CNN, we achieve the final accuracy of classification reported as 99.45%. This is higher than the previous reported research [11]. Table 3 shows the result comparisons.

Fig. 9 represents the accuracy and Fig. 10 shows the loss curves for the model. The blue curve shows the training accuracy and the orange is for the validation. Epochs in the image means the number of times the model has been tested on the data. These graph images have been generated by the same code as that is used to train the model. From Fig. 9 it is seen that after seven epochs the accuracy and validation accuracy of the model are closely following the same curve.

Fig. 10 shows the loss curve of the model developed here. The loss data reported during the training phase of the research is plotted using "matplotlib". Over fitting is a problem in the neural networks that results in irregular accuracy when training the same. Normally, when training accuracy increases, validation accuracy should also increase. When the opposite happens i.e. more training and less validation accuracy it indicates that the model is overfitting. When both the accuracy curves are very close to each other, the model is said to be accurate.

Fig. 11 represents the confusion matrices of the model developed in this research. The confusion matrix gives the prediction performance of a model individually per class – 'Center':0, 'Donut':1, 'Edge-Loc':2, 'Edge-Ring':3, 'Loc':4, 'Random':5, 'Scratch':6, 'Near-full':7, 'none':8. If the number on the highlighted diagonal is high, the accuracy of the model for that class is high.

Fig. 11 also represents the normalized confusion matrix of the model. In order to better represent the data in the confusion matrix, we represent the true predictions of the model per class respectively. The normalized confusion matrix represents the per class accuracy of the model.

6. Conclusion

Accuracy in the detection of semiconductor failures is very important to reduce the waste. The demand of semiconductors is ever increasing, and the rare earth elements are in a fixed amount on the planet. Once the rare earth metals have melted into a glass (semiconductor), it is not possible to recycle them. In contrast to the previous works, using the deep CNN and DNN has been used here. This gives a boost in accuracy performance of the neural net-

work developed (Table 3). This finally helps to reduce waste in the semiconductor fabrication process. Since the accuracy of the model is very high, until more data is collected, further progress is not possible anymore.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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